WORKED SOLUTION

STEP II 1999, Question 10

Mechanics • Successive Collisions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q10

The key idea

Every impact in this chain looks the same: a particle of mass μ moving with speed u hits a particle of mass $q\mu$ at rest. So the first-impact calculation, scaled, describes *every* impact. Once we know each particle's final velocity, the question of whether any extra impacts happen reduces to comparing neighbouring final velocities, and with $q = e$ the velocities become so simple that the energy loss is just a geometric series.

Solution

The first impact

Model the particles as point masses moving along the line, and take the direction from P_1 towards P_2 as positive. Let v_1 and v_2 be the velocities of P_1 and P_2 just after the impact. Conservation of momentum and Newton's law of restitution give

$$\begin{aligned} mV &= mv_1 + qm\,v_2, \\ v_2 - v_1 &= eV. \end{aligned}$$

Substituting $v_2 = v_1 + eV$ into the first equation gives $V = v_1 + qv_1 + qeV$, so

$$v_1 = \left(\frac{1-eq}{1+q}\right)V, \qquad v_2 = v_1 + eV = \frac{1-eq+e+eq}{1+q}\,V = \left(\frac{1+e}{1+q}\right)V,$$

as required.

Every impact has the same shape

Write

$$\alpha = \frac{1-eq}{1+q}, \qquad \beta = \frac{1+e}{1+q}.$$

Nothing in the calculation above used the actual value of m: a particle of mass μ moving at u that hits a particle of mass $q\mu$ at rest leaves with velocity αu and sends the other off

with velocity βu. Since each P_{k+1} has q times the mass of P_k, this applies to an impact of P_k on P_{k+1} whenever P_{k+1} is still at rest.

Exactly $N-1$ impacts when $q \le e$

Suppose $q \le e$. Then $eq \le e^2 < 1$, so $\alpha > 0$; also $\beta > 0$.

Follow the chain. P_1 hits P_2, leaving P_1 with velocity αV and P_2 with βV. Since $\beta V > 0$, P_2 moves towards P_3, which is at rest, and hits it; this leaves P_2 with $\alpha\beta V$ and P_3 with $\beta^2 V$. Continuing, P_k arrives at the stationary P_{k+1} with velocity $u_k = \beta^{k-1}V > 0$, so there is one impact between each neighbouring pair: at least $N-1$ impacts. After them the velocities are

$$w_k = \alpha\beta^{k-1}V \quad (1 \le k \le N-1), \qquad w_N = \beta^{N-1}V.$$

All of these are positive, so everything moves the same way.

Ask yourself How do we know P_k never catches P_{k+1} again?

After their impact, P_k moves at constant velocity w_k from then on, because its only possible later contacts are with P_{k+1} or P_{k-1}. Meanwhile P_{k+1} moves at $u_{k+1} = \beta^k V$ until it hits P_{k+2}, and at w_{k+1} afterwards (or for ever, if $k+1 = N$). The gap between P_k and P_{k+1} can only close if P_k is moving faster, and:

- $u_{k+1} > w_k$, since $\beta^k V > \alpha\beta^{k-1}V$ is equivalent to $\beta > \alpha$, i.e. $1+e > 1-eq$, which is true;
- for $k \le N-2$, $w_{k+1} \ge w_k$, since $w_{k+1}/w_k = \beta = \dfrac{1+e}{1+q} \ge 1$ exactly because $q \le e$;
- for $k = N-1$, $w_N > w_{N-1}$, since $w_N/w_{N-1} = \beta/\alpha > 1$.

So P_{k+1} is never slower than P_k after their impact, and the gap between them never shrinks. (When $q = e$ we get $w_{k+1} = w_k$: the gap stays constant, but it doesn't close.) This holds for every neighbouring pair, so no further impacts occur and there are *exactly* $N-1$ impacts.

This is where $q \le e$ matters. If $q > e$ then $\beta < 1$, each particle ends up slower than the one behind it, and the chain catches up with itself, producing more impacts.

Kinetic energy lost when $q = e$

Now $q = e$, so

$$\beta = \frac{1+e}{1+e} = 1, \qquad \alpha = \frac{1-e^2}{1+e} = 1-e.$$

Every particle is hit at speed V; afterwards $P_1, \dots, P_{N-1}$ move at $(1-e)V$ and P_N moves at V. With P_k of mass $e^{k-1}m$, the final kinetic energy is

$$\begin{aligned} T &= \sum_{k=1}^{N-1} \tfrac{1}{2} e^{k-1} m (1-e)^2 V^2 + \tfrac{1}{2} e^{N-1} m V^2 \\ &= \tfrac{1}{2} m V^2 \left[(1-e)^2 \cdot \frac{1-e^{N-1}}{1-e} + e^{N-1} \right] \\ &= \tfrac{1}{2} m V^2 \left[(1-e)(1-e^{N-1}) + e^{N-1} \right], \end{aligned}$$

summing the geometric series $1 + e + \cdots + e^{N-2} = \dfrac{1-e^{N-1}}{1-e}$. The initial kinetic energy is $\frac{1}{2}mV^2$, so the loss is

$$\begin{aligned} \tfrac{1}{2} m V^2 - T &= \tfrac{1}{2} m V^2 \left[(1-e^{N-1}) - (1-e)(1-e^{N-1}) \right] \\ &= \tfrac{1}{2} m e \left(1 - e^{N-1} \right) V^2, \end{aligned}$$

which is the required result.

Checks

- *Impact by impact.* When $q = e$, the impact of P_k on P_{k+1} takes P_k from V to $(1-e)V$ and P_{k+1} from 0 to V, losing $\frac{1}{2}e^{k-1}mV^2\left[1-(1-e)^2-e\right] = \frac{1}{2}m(1-e)e^kV^2$. Summing over $k = 1, \dots, N-1$ gives $\frac{1}{2}m(1-e)V^2 \cdot e\,\dfrac{1-e^{N-1}}{1-e}$, the same answer.
- *Limits.* As $e \to 0$ the loss tends to 0, and indeed with $q = e \to 0$ the later particles have negligible mass. With $N = 2$ the formula gives $\frac{1}{2}me(1-e)V^2$, which matches a direct calculation for a single impact.
- *Numerically.* We simulated the motion event by event for random $N \le 12$, random gaps and random $q \le e$, and always found exactly $N-1$ impacts. For $N = 5$, $q = e = \frac{1}{2}$, $m = V = 1$ the simulated loss was $0.234375 = \frac{15}{64}$, in agreement with $\frac{1}{2} \cdot \frac{1}{2}\left(1 - \frac{1}{16}\right)$.

Common slips

- Checking only that the first $N-1$ impacts happen. "Exactly" also needs an argument that no particle catches the one in front later on.
- Comparing P_k's final velocity with P_{k+1}'s velocity *just after* their impact (βu_k), rather than with its final velocity after it has hit P_{k+2} and slowed down.
- Getting the sign wrong in the restitution law: it is the speed of separation, $v_2 - v_1$, that equals e times the speed of approach.

Beyond STEP

A chain of masses that pass momentum along one at a time is a one-dimensional model of a shock moving through a granular material. Chains of steadily decreasing masses, like this one when $q < 1$, are studied by physicists as a way to spread out and absorb impacts. A nice way to reorganise the energy calculation is through the *reduced mass* $\mu = m_1 m_2/(m_1 + m_2)$: one impact loses $\frac{1}{2}\mu(1 - e^2)w^2$, where w is the speed of approach. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops collisions in the centre-of-mass frame, where results like this become transparent.

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